##Focus on food order and food delivery service only OR
focus on product order and delivery of product only.



FACULTY OF COMPUTING
UTM JOHOR BAHRU

SCST1143 - DATABASE
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Project: Phase 1
Project Proposal & Database Requirement

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1.0 Introduction

Online shopping and food delivery have been used by many people. This has become an important part of our daily life as it is easier to buy necessities in life. Nowadays, it is easy to buy almost anything or order food from anywhere with just a few taps on our phone. Students and staff at University Technology Malaysia, Johor Bahru often use messaging apps such as Telegram and WhatsApp to promote their food menus and parcel delivery services. However, these promotions usually get mixed in with other unrelated messages like club updates, official announcements, and general discussions. This makes it hard for students to keep track of important promotions and delivery services which causes them to miss out on great offers and services available on campus.

Therefore, we want to create a system that organizes these promotions and delivery services in one convenient place. The system would act as a virtual marketplace where sellers can share their menus, deals, and delivery options without getting mixed up with other messages. With this way, students can easily discover ongoing offers and parcel delivery services that they want without having to deal with many chats in the group. This is very helpful for those who live far from the parcel pick-up points or do not having any transportation.

Furthermore, the system would not only centralize promotions but also provide features like real-time notifications for special promotions and an organized menu for easy browsing. Overall, this system aims to create a smoother and more efficient experience for both sellers and buyers at UTM.

2.0 Background Study

Based on our research and observations, most students sell food and make delivery services through messaging apps. For instance, students offer the menu of food they want to sell through Telegram and Whatsapp applications where the application has many members. This can attract many people to buy the food. Unfortunately, sometimes the message does not reach others.

As a student of UTM, we think that this method of promotion is inefficient. For example, students need to scroll all the way to the messages that were sent to make an order. This causes students to be lazy to make an order which results in losses to the seller who promotes their menus in the group chat. Not only that. It is a waste of time to send menus everyday in the chat group and treat each person who orders by personal message.

For parcel delivery, some students need their item in the near future. Basically, they need to wait for the parcel delivery service to pop out in the group chat so they can make a request to ask the rider to pick up their items. If the service does not pop out that time, they have to take the parcel themselves which will take time and need to find transport.

With the current method, it clearly shows that sellers and students need to improve sales and service methods. This needs to be done to ensure that the buying and selling that takes place can be done easily and run smoothly. If this happens, we can avoid such problems like taking the wrong order, having to stand in long queues to pick up parcels and many more.

3.0 Problem Statement

Inefficient promotion method

Currently, messaging apps are being used by the sellers to promote their foods and products in the campus. Although it may sound convenient, it's actually quite inefficient for both sellers and students. Most students may frequently miss out on important promotions and updates from the sellers since they are easily overlooked as they get mixed up with other messages especially in public group chats such as college and faculty groups. Some promotion messages can also be wrongly accused of scams.

Inconsistent delivery time

Most e-commerce apps nowadays take a few days to process their delivery. This process also depends on the sellers and the company parcel distribution center. Location of the sellers can also contribute to such problems that lead to inconsistent delivery time. Sometimes, the products can also be delivered to the wrong address which results in more days needed to deliver. Students who require the products very soon will have difficulties waiting to receive their products.

High cost of delivery

Although buying things online outside the campus can be quite cheap and convenient, the final price can be totaled up to more than reasonable price due to delivery fees set by the system. Amount of delivery fees can vary depending on the courier used to deliver the products and location of the sellers, especially if they are from other countries such as China. Students who are new to the system can make a mistake by overpaying the product due to not checking the location.

4.0 Proposed Solutions

After some discussions and meetings with the group members, we decided to invent a system that requires a website and database management system called UTMart. This system will allow everyone in the campus including the sellers to easily access, promote, buy and sell products and foods with each other. Staff play an important role in managing the database system to ensure more secure and reliable information kept accordingly.

First of all, the website will be provided with user-friendly interface design and features that can be easily used by the students. They are required to sign in their details such as username, full name, identification number/passport number, nationality, birthdate and more. Other users such as local sellers are also required to sign in to provide more secure interaction with every user.

Other than that, the website will allow sellers to create a section for their own business to promote and sell their products. The section can also be filled with information such as details of the products, working hours and their location. This will allow users to easily find their products based on their likings. Students can also put up a rating in the range of 1-5 stars based on their experience.

Furthermore, the system will connect everyone inside the campus. Students who are interested in selling their products can also register as sellers to increase their savings throughout their semester. This feature will allow students to help each other as they don't require a few days to deliver and process their deliveries. As a result, students who are in dire need of the products can use them very soon. Those who only need it for short periods of time can choose the rent option to reduce the price.

In addition, since the system is being developed around the campus, this will allow for reasonable delivery fees for students. The system will calculate based on the seller location and the student location for a more consistent delivery fee. The delivery time will be also short and doesn't require external courier service. While registering into the system, students are required to state their college which will eliminate the chance of the products being delivered to the wrong address.

To provide a systematic payment method for user to make payment of their order.

5.0 Objectives

To develop an online system for purchasing products and delivery service.

- **Create a Centralized System:** Design and develop an integrated and centralized system that allows sellers and students at UTM to buy and sell items and food easily.
- **Organized Categories**: Establish organized categories for products and food to ensure a seamless and quick browsing experience for students, making it easier for them to find what they need.
- **Ensure Transparency:** Provide a clear and transparent transaction system between sellers and students, which increases trust and encourages engagement.
- **Comprehensive Rating System:** Enable users to rate the service provided by sellers using a rating system from 1 to 5 stars, facilitating decision-making for students.
- **Order Tracking System:** Develop a system that allows students to track the status of their purchased items and meals, along with reminders for rented products to ensure timely returns.

To develop a tracking module for helping users to trace their order and parcel.

To generate the required report such as sales report, order report, payment report, etc.

6.0 Scope of the project

explain about the limitation of your proposed system (in point forms)

This project will be on a website. Users will have to register an account by typing in their matric number and password. After registration, the user is required to complete their details such as their name, birthdate, email address, nationality and more. When the user logs in, they will be presented with different interfaces according to their respective categories. Our system will identify each user and present them the corresponding interface based on the categories of users they are in.

Customers Interface

Customer of the system is the student or staff of Universiti Teknologi Malaysia. Customers will be presented with three buttons with each being categories of products they are searching for. The buttons being items, food and rented items.

If customers click on the 'items' category, they will be presented with items that are available for sale. When they click on the product, they will be presented with details of the item such as the price and the condition of the item. In the same page of the chosen item, it will also provide the contact details of the seller of the item, the working hour and the location of the seller.

If customers click on the 'food' category, customers will first be shown the list of restaurants in Universiti Teknologi Malaysia. After the customer has chosen their selected

restaurant, customers are presented with the contact details of the restaurant, the working hours of the restaurant and the menu of the restaurant. When a menu is picked, students can leave a note to the restaurant through a comment box to customize their order such as to customize the level of spiciness or include or exclude a certain ingredient from their food.

Customers are able to place orders for the item. Before the order is placed, customers are required to confirm their order by clicking the 'Order' button. After that, customers will be given an option to ask for delivery or self-pickup their order. After choosing the order, the customers are required to choose to pay by cash on delivery (COD) or by DuitNow payment. Once an order is placed, customers are not allowed to cancel their order. If the customer chose to ask for delivery, students will be given the drivers information to communicate with the driver for their delivery. After the item or food is delivered, customers can rate the item or food received from 1 to 5 points with 1 being terrible and 5 being excellent.

If a customer clicks on the 'rented item' button, they will be presented with items that are available for rent. When customers click on a rental item, they can book the item rented. Customers are required to fill in the duration of their rent by entering the start of rent and end of rent on the calendar provided by our system. Once rented, students will be presented with time duration for their rented item. When the end duration is near, student will be given push notification as a reminder

Seller interface

Sellers of the system can be anyone who owns a business in Universiti Teknologi Malaysia such as food vendor, restaurant owner, student or staff member. Sellers are required to upload their banking information such as their bank account number and DuitNow QR code for the customer to pay. Sellers will be given three categories of product which are 'item', 'food' and 'rented items'.

If the Seller chose the 'item' category, sellers are required to upload the picture of the item, price, condition and other necessary information of the item being sold. If the seller choose the 'food' category, seller are required to upload the picture of the food, price, ingredients and other necessary details of the food.If the seller chose the 'rented item' category, seller are required to upload the picture of the rented item, price per unit time and other necessary details of the item. Sellers can edit their page, delete or change the details of their product being sold. When the seller receives an order, they will be able to see the profile of the customer who is ordering their product. Sellers are also able to view the ratings given by the customer. Sellers can look for a driver to deliver their items to the customer.

Driver interface

Drivers can be anyone with transportation in Universiti Teknologi Malaysia. Drivers are given a choice to choose whether to deliver ‘item’ or ‘food’. If the driver chose to deliver ‘item’, they may only deliver ‘item’ only. Driver who chooses to deliver an item can deliver up to 5 items in 1 time. After all the items are delivered, they can choose to deliver items again. If the driver chose to deliver ‘food’, they may only deliver ‘food’ only. Driver who chooses to deliver food can deliver up to 3 food in 1 time. After all the food is delivered, they can choose to deliver items again. Driver can view the location of the customer before they accept the order. Drivers will be redirected to google map to navigate the location of their customer. Once an order has been fulfilled, the driver must confirm the delivery is a success by clicking ‘Order delivered’.

7.0 Project Planning

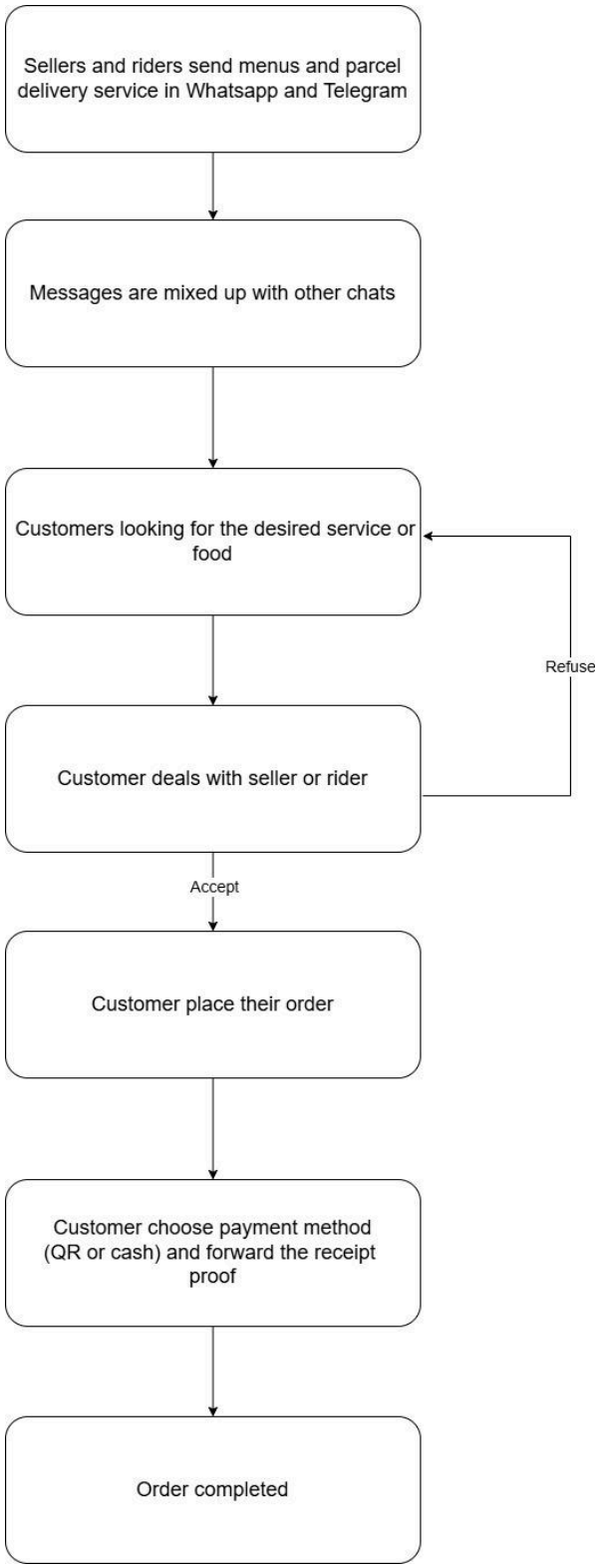
No	Activity	Person In Charge	Start Date	Finish Date
1	First meeting with group members	Jeremy	20/10/24	20/10/24
2	Each member propose and finish module	Fikri	20/10/24	22/10/24
3	Task distribution	Hafiz	23/10/24	23/10/24
4	Finish proposal	Mohamed	23/10/24	31/10/24



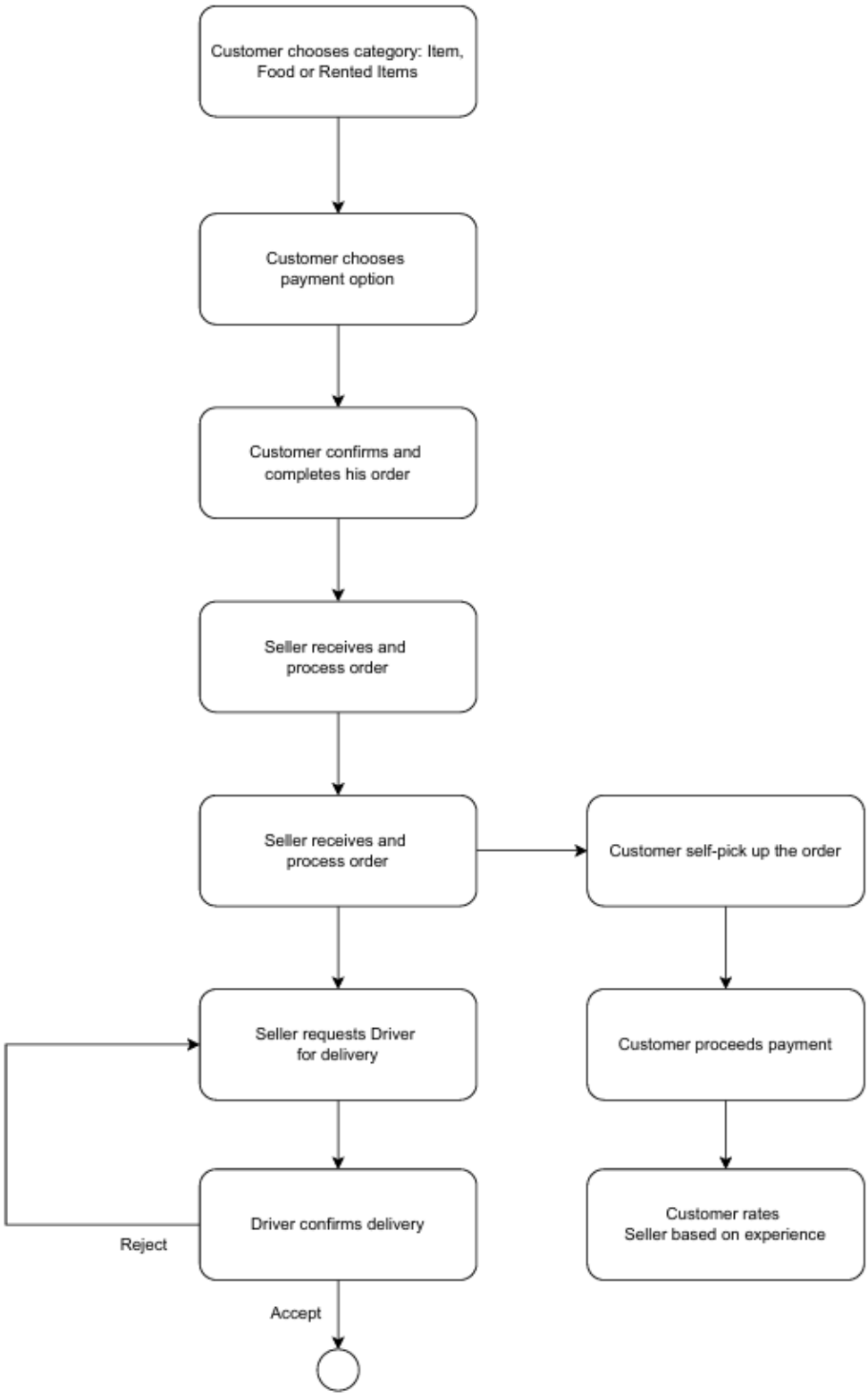
add more activities such as design conceptual database, design logical database design, conduct interview, distribute questionnaire, prepare report writing, etc.

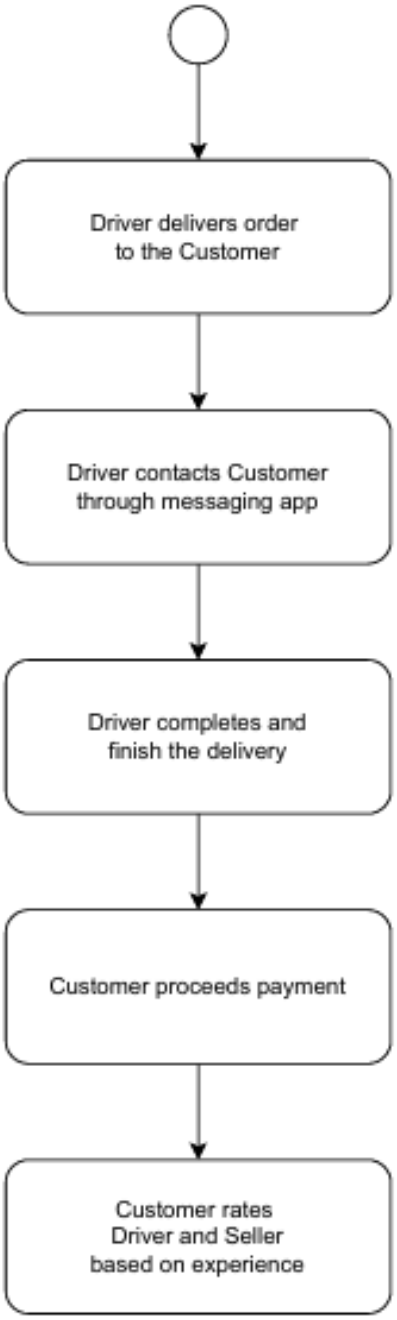
8.0 Requirement Analysis

8.1 Current Business Process



8.2 Proposed Business Process




Driver delivers order
to the Customer
Driver contacts Customer
through messaging app
Driver completes and
finish the delivery
Customer proceeds payment
Customer rates
Driver and Seller
based on experience

9.0 Benefit and Summary of Proposed System

- C**entralized Platform:** UTMart will offer a centralized platform for students and staff to buy and sell products and food within the UTM campus. This eliminates the clutter of group chats, making it easier for users to discover available services and deals.

- **User-Friendly Interface:** The system will provide a clean, intuitive interface that allows sellers to list their products, and buyers to browse organized categories. This will streamline the process of making transactions and improve user satisfaction.

- **Real-Time Notifications:** Students and staff will receive real-time notifications about new promotions, delivery statuses, and special offers, ensuring that they never miss out on important updates.

- **Fair Delivery System:** The system will calculate delivery fees based on the distance between the seller and the buyer, ensuring that delivery charges remain fair and consistent within the campus.

- **Improved Transaction Transparency:** UTMart will feature a secure transaction system that includes verification of buyer and seller information, as well as tracking of delivery orders. This will enhance trust between users.

- **Rating and Review System:** Buyers will have the ability to rate sellers based on their experience, which will provide transparency and accountability, helping future users make informed decisions.

- **Efficient Delivery Process:** Unlike external e-commerce platforms that may take days to deliver, UTMart's system ensures fast delivery within the campus, avoiding delays and making it convenient for students and staff who need items urgently.

In conclusion, UTMart aims to resolve the inefficiencies in the current system by providing a streamlined, user-friendly platform tailored specifically to the UTM community. This system not only enhances the buying and selling experience but also fosters a more connected and efficient campus marketplace.